Day 3 LAB: Build, copy, and read a gene

- Today's target: a gene's message is copied, and the copy is read to build a chain of amino acids (LS1-1.3)

The idea: copy, then read

- DNA stays safe in the nucleus.
- The cell COPIES the gene's message from the template strand.
- The COPY is read three letters at a time.
- Each triplet gives one amino acid. The amino acids make a chain.

Base key and code table

- Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Safety

- Do not eat lab materials until your teacher says you may.
- Toothpicks are sharp.
- Tell your teacher about any food allergy.
- Clean up your table and wash your hands.

Part A and B (14 min)

- A: build the template strand T A C C C T G A C A A A
- B: build the partner strand (A with T, C with G) and make the ladder.
- Have your teacher check the ladder.

Part C (10 min)

- Pull the strands apart. Keep the template.
- Build the COPY by pairing with the template.
- Read the copy three letters at a time with the code table.
- Write the amino acid chain.

Part D (7 min): change one letter

- Change the 4th base of the template: it becomes T A C T C T G A C A A A
- Build the new copy and read it.
- How many amino acids changed? Which one?
- Predict: could this protein work differently?

Wrap up

- Connect questions: what did one letter do? How is the model different from real DNA?
- Clean up.
- Exit ticket: CFA LS1-1.3 (6 minutes).